Effective-sample-size calibration of a modality test for cluster homogeneity and the number of classes in spatially autocorrelated data

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Abstract

Selecting the number of clusters $\boldsymbol{K}$ is a basic step in unsupervised classification. The standard internal criteria (the gap statistic, the average silhouette, the Calinski–Harabasz and Davies–Bouldin indices) assume independent observations, and so are miscalibrated on spatial data. There, neighbouring locations are autocorrelated, the effective sample size is far below the nominal one, and a smooth large-scale trend mimics discrete structure. We show that the consequence is systematic over-detection: on autocorrelated fields without discrete classes these criteria report spurious clusters in every one of **75** controlled realisations, and on real imagery they fragment single land-cover regions. We recast the choice of $\boldsymbol{K}$ as a recursive homogeneity test: at each step, whether a region is a single class or carries multimodality in excess of spatial autocorrelation. The proposed method, ESS-Dip, evaluates the Hartigan dip statistic along the locally optimal split direction. It calibrates the statistic against the effective sample size derived from a within-class autocorrelation range, which a local declustering estimator separates from the between-class scale; a low-order trend is removed adaptively. A recursive partitioning returns $\boldsymbol{K}$, and the test carries an asymptotic false-split-control property. On a factorial simulation with known ground truth, ESS-Dip reaches unit specificity with competitive recovery; its recall-oriented variant ESS-Dip-R, calibrated against the model's Gaussian null, attains the best balanced score of fifteen criteria at the same specificity. On the Indian Pines, Salinas and Sentinel-2 scenes both identify single land-cover regions as homogeneous where the standard criteria do not. The method is conservative by design, and its negatives are reliable: when it reports a single class that verdict can be trusted, suiting ESS-Dip to use as a homogeneity guard or a stopping rule rather than as an exhaustive enumerator of weakly separated or strongly imbalanced classes, a trade-off we characterise. Code and data are released.

Keywords: number of clusters, cluster validity, spatial autocorrelation, effective sample size, declustering, unsupervised classification, remote sensing

1 Introduction

Unsupervised classification partitions a data set into groups without labelled examples, and in doing so it must answer a prior question: how many groups are there? The choice of the number of clusters K governs every downstream interpretation, and in the analysis of remotely sensed imagery, where K is the number of land-cover classes an analyst will map, it is made routinely and consequentially. A mature toolkit exists to make it: the gap statistic (Tibshirani et al. 2001), the average silhouette (Rousseeuw 1987), the Calinski–Harabasz (Caliński and Harabasz 1974) and Davies–Bouldin (Davies and Bouldin 1979) indices each score a partition by its internal geometry and select the K that optimises the score.

These criteria share a hidden assumption: that the observations are independent. Their reference levels, the uniform null of the gap statistic and the implicit baseline of a silhouette, describe what the geometry would look like were the samples exchangeable draws. Spatial data are not exchangeable. Neighbouring pixels are autocorrelated (Tobler 1970), so the effective number of independent observations is far below the nominal pixel count (Cressie 1993), and a smooth illumination or topographic gradient spreads the feature cloud along a manifold that an independence null cannot distinguish from genuine groups. The consequence, which we quantify in Sections 4–5, is systematic *over-detection.* On synthetic fields containing no discrete classes the gap statistic reports on average between seven and eight; on real imagery the silhouette, Calinski–Harabasz and Davies–Bouldin indices fragment a single, homogeneous crop field into several sub-classes in every instance we examined. A criterion that cannot recognise the absence of clusters is a precarious foundation for deciding their number.

Correcting this is not a matter of swapping in a better null. Because the within-cluster dispersion that these criteria minimise is a second-order quantity, a reference matched to the data's second-order spatial structure, its variogram, reproduces the same dispersion and explains the clusters away. Distinguishing real classes from autocorrelation therefore requires information beyond second order, namely the *multimodality* of the feature distribution; and distinguishing real classes from a smooth trend requires separating a continuous drift from discrete structure, a separation the geostatistical decomposition into trend and residual makes natural. The idea of calibrating a clustering criterion against an autocorrelation-aware null is itself not new. Hennig and Lin (2015) do exactly this for discrete presence–absence data, but it has not been carried to the continuous-field setting of gridded imagery, nor has the confounding of trend with class been addressed in selecting K.

This paper supplies both. We recast the question *how many classes?* as a sequence of homogeneity decisions: at each node we test whether a region is a single class or instead carries *multimodality in excess of spatial autocorrelation*, and make that test valid on autocorrelated fields by calibrating it at the effective sample size. The construction rests on three geostatistical devices: the effective sample size itself (Cressie 1993; Dutilleul 1993), evaluated at the indicator integral range appropriate to a functional of the distribution rather than the field integral range that governs the mean (Section 3.7), which sets the calibration; a *local* estimate of the within-class autocorrelation range, in the manner of declustering (Isaaks and Srivastava 1989; Deutsch and Journel 1998), which prevents the between-class mosaic from corrupting that calibration; and an adaptive trend–residual decomposition, which removes a smooth drift only when it is statistically resolvable. A recursive partitioning driven by the calibrated dip statistic (Hartigan and Hartigan 1985) returns K. Because the test is tuned for specificity, its negatives are trustworthy: when it declares a region homogeneous that verdict can be relied on, which makes ESS-Dip as useful as a homogeneity guard or a stopping rule for region-growing as it is as a conservative estimator of K. Our contributions are:

- We document, on controlled ground truth, that the standard number-of-clusters criteria over-detect under spatial autocorrelation without exception, failing to recognise class-free scenes in every one of 75 realisations (Section 4).
- We propose *ESS-Dip*, an effective-sample-size-calibrated test of cluster homogeneity (whether a region holds discrete structure beyond its spatial autocorrelation), combining a within-class range from local range estimation with adaptive trend removal (Section 3); applied recursively it returns a conservative K, and we state an asymptotic false-split-control property (Proposition 1) that the method confirms empirically; a recall-oriented variant, ESS-Dip-R, trades the distribution-free uniform null for the model's Gaussian null to recover more structure at the same specificity.
- We evaluate fifteen criteria on a factorial simulation with known K, including the calibrated-validity-index method of Hennig and Lin (2015) that ESS-Dip generalises; ESS-Dip reaches unit specificity, its recall variant ESS-Dip-R attains the best balanced score at the same specificity, and an ablation attributes the gain to the local range and the adaptive detrending (Section 4).
- We confirm the behaviour on three real scenes spanning hyperspectral and multispectral sensing (Indian Pines, Salinas, and a Sentinel-2 scene labelled by ESA WorldCover), where ESS-Dip identifies single land-cover regions as homogeneous where the standard criteria do not (Section 5).

The remainder of the paper is organised as follows. Section 2 reviews number-of-clusters criteria, the geostatistical notions of effective sample size and declustering, and modality-based clustering, and positions the contribution. Section 3 develops the method and its calibration. Sections 4 and 5 report the simulation study and the application to land-cover imagery. Section 6 discusses the precision–recall trade-off and limitations, and Section 7 concludes. Code and data are released to reproduce every result.

2 Background and related work

2.1 Selecting the number of clusters

Choosing the number of clusters K is a classical problem in unsupervised analysis, and a large family of *internal* criteria address it by scoring a partition against the geometry of the data alone. The gap statistic (Tibshirani et al. 2001) compares the within-cluster dispersion to its expectation under a reference distribution with no clustering, typically uniform over the data's bounding box; the average silhouette (Rousseeuw 1987), the Calinski–Harabasz ratio (Caliński and Harabasz 1974) and the Davies–Bouldin index (Davies and Bouldin 1979) balance within- against between-cluster scatter. Common to all is the assumption that the observations are exchangeable: the reference distribution, or the index's implicit baseline, treats the samples as independent draws. When this holds the criteria are well calibrated; when it fails they are not. Spatial data are the archetypal failure, because nearby locations are correlated: "everything is related to everything else, but near things are more related than distant things" (Tobler 1970).

2.2 Accounting for spatial dependence

That an independence reference over-detects structure in dependent data is known across several fields. In spatial epidemiology, Goovaerts and Jacquez (2004) show that a complete-spatial-randomness null over-identifies disease clusters and propose, in its place, a geostatistical *spatial neutral model* (realisations reproducing the histogram and variogram of the data by sequential Gaussian simulation), against which far fewer locations remain significant. In neuroimaging, Eklund et al. (2016) trace inflated false-positive rates in cluster-extent inference to a mis-specified parametric model of the spatial autocorrelation. The common remedy is a null that *preserves* the second-order spatial structure while removing the effect being tested, realised either by geostatistical simulation (Goovaerts and Jacquez 2004) or by spatially constrained randomisation such as Moran spectral randomization (Wagner and Dray 2015).

The same logic has been brought to bear on K itself. Hennig and Lin (2015) calibrate a validity index (the gap, silhouette or prediction strength) against its distribution under a problem-specific null that encodes non-clustering structure, including spatial autocorrelation, and use the calibrated index both to test for clustering and to estimate K; on a biogeographic example a plain Gaussian null favours eight clusters while the autocorrelation-aware null reduces this to two and renders the apparent clustering non-significant. Their framework is the conceptual basis of the present work. It is, however, formulated for discrete presence–absence data with an algorithmic disjunction-parameter null, and addresses neither the *continuous-field* setting of gridded imagery nor the confounding of a smooth large-scale *trend* with discrete classes. The present contribution supplies exactly these: a Gaussian-random-field null on the lattice, a within-class range obtained by local range estimation, an adaptive trend–residual separation, and an effective-sample-size calibration of a modality statistic.

The quantitative device that makes this possible is the *effective sample size*: under spatial autocorrelation the variance of a sample statistic behaves as if computed from

$n_{\text{eff}} < n$ independent observations, a reduction governed by the integral of the autocorrelation (Cressie 1993). Dutilleul (1993) operationalised this idea to correct the degrees of freedom of a correlation test between two spatial processes; we use it to recalibrate, rather than to correct after the fact, the reference distribution of a clustering criterion. The autocorrelation scale it requires is the variogram range (Matheron 1963; Cressie 1993), and the safeguard against letting large homogeneous regions dominate its estimate is *declustering*, a standard geostatistical correction for spatially clustered support (Isaaks and Srivastava 1989; Deutsch and Journel 1998).

2.3 Modality as a clustering signal

Because k-means dispersion is itself a second-order quantity, a null matched to the data's second-order structure can explain away the very clusters one seeks; a criterion that survives such a null must use information beyond second order. Multimodality is the natural choice. The dip test of Hartigan and Hartigan (1985) measures the departure of a univariate sample from unimodality, and dip-means (Kalogeratos and Likas 2012) turns it into a divisive procedure that splits a cluster only when a projection of it is significantly multimodal, thereby estimating K. These methods assume independent observations; our contribution is to make the dip test valid under spatial autocorrelation by calibrating it at the effective sample size, and to pair it with the trend removal that prevents a smooth gradient from being read as two modes.

2.4 Distinction from spatially constrained clustering

A separate line of work *imposes* spatial contiguity on the partition: regionalization methods such as SKATER (Assunção et al. 2006), ClustGeo (Chavent et al. 2018) and the max-p-regions problem (Duque et al. 2012) constrain clusters to be spatially connected. These address a different question, how to partition space into contiguous regions, and still require the number of regions as an input or a tuning parameter. We do not constrain the partition; we correct the criterion that selects *how many* groups are present, and the two are complementary: the effective-sample-size calibration developed here could in principle inform the stopping rule of a regionalization just as it informs the recursion of Section 3.

3 Mathematical formulation

We formulate the selection of the number of clusters as a sequence of hypothesis tests for *multimodality in excess of spatial autocorrelation*. Section 3.1 fixes notation; Section 3.2 states the null against which a candidate split is judged; Sections 3.3–3.6 develop the test statistic, its effective-sample-size calibration, the within-class range estimator, and the trend–residual decomposition; Section 3.8 assembles these into a recursive partitioning algorithm; Section 3.9 states the resulting false-split control; and Section 3.10 discusses computation.

3.1 Setting and notation

Let $D \subset \mathbb{Z}^2$ be a finite regular lattice (the image domain) with $N = |D|$ pixels, and let

$$\boldsymbol{z} : D \to \mathbb{R}^p, \qquad \boldsymbol{z}(s) = \big(z_1(s), \dots, z_p(s)\big)^\top,$$

be a multivariate field assigning to each site $s \in D$ a p-dimensional feature (e.g. a spectral signature). Writing $\boldsymbol{z}$ as N feature vectors $\{\boldsymbol{z}(s)\}_{s \in D}$, unsupervised classification seeks a partition of D into K groups whose feature vectors are internally homogeneous. Our object of interest is the *number* of groups K, not the partition itself.

We treat $\boldsymbol{z}$ as one realisation of a second-order stationary random field and decompose it, in the geostatistical manner, into a deterministic trend and a stochastic residual,

$$\boldsymbol{z}(s) = \boldsymbol{m}(s) + \boldsymbol{\varepsilon}(s), \tag{1}$$

where $\boldsymbol{m}$ is a smooth large-scale drift (illumination, topography, atmospheric gradients) and $\boldsymbol{\varepsilon}$ is a zero-mean residual carrying the short-range autocorrelation and any discrete (between-class) structure. The second-order dependence of $\boldsymbol{\varepsilon}$ is summarised by its correlogram $\rho(\boldsymbol{h})$, or equivalently its variogram, and condensed into a single scalar by the *decorrelation area*, the integral range of Cressie (1993),

$$a = \iint \rho(\boldsymbol{h})\, d\boldsymbol{h}, \tag{2}$$

the area occupied by one spatially independent observation. Estimating a from a fitted variogram is the subject of Section 3.5; for a subset $C \subseteq D$ we write $n_C = |C|$.

3.2 Clusters as excess multimodality

Internal validity criteria for K, namely the gap statistic (Tibshirani et al. 2001), the average silhouette, Calinski–Harabasz and Davies–Bouldin, compare a within-cluster dispersion against the value expected with *no* clustering, under an *independent* reference distribution. Two features of spatial data invalidate this comparison. First, autocorrelation reduces the number of independent observations far below N, so dispersion-based statistics are evaluated as if the sample were much larger than its information content (Cressie 1993; Dutilleul 1993). Second, a smooth drift $\boldsymbol{m}$ spreads the feature cloud along a low-dimensional manifold and is read as structure by an independence null. The combined effect is a systematic *over-detection* of clusters, which we document empirically in Section 4.

We therefore replace the dispersion comparison by a test of *modality.* Discrete classes manifest as a *multimodal* feature distribution; a single class (however autocorrelated, and however smoothly it varies in space) is *unimodal.* This motivates the reference hypothesis under which a candidate cluster is judged homogeneous.

Assumption 1 (Unimodal Gaussian-random-field null)**.** Under the null H_0 a cluster's residual field $\boldsymbol{\varepsilon}$ restricted to C is a second-order stationary Gaussian random field with decorrelation area a and *no* discrete component. In particular the marginal law of any fixed linear projection $\boldsymbol{u}^\top \boldsymbol{\varepsilon}$, $\boldsymbol{u} \in \mathbb{R}^p$, is univariate Gaussian, hence unimodal.

Assumption 1 is the continuous-field counterpart of the autocorrelation-aware null of Hennig and Lin (2015), who calibrated a validity index against a non-clustering reference for point data; here the reference is a Gaussian random field, the maximum-entropy unimodal model given the second-order structure, of the kind routinely generated by sequential Gaussian simulation (Goovaerts and Jacquez 2004; Wagner and Dray 2015).

3.3 The dip statistic along the split direction

Modality is quantified by Hartigan's dip (Hartigan and Hartigan 1985). For a univariate sample with empirical distribution function F_m (size m), the dip is the supremum distance to the closest unimodal distribution,

$$\operatorname{dip}(F_m) = \inf_{G \in \mathcal{U}} \sup_{x \in \mathbb{R}} |F_m(x) - G(x)|, \tag{3}$$

$\mathcal{U}$ being the class of unimodal distribution functions. The dip is 0 for a perfectly unimodal sample and grows with the separation between modes.

Testing (3) on a one-dimensional projection requires a direction along which a genuine bimodality, if present, is exposed. Rather than the leading principal axis, which can hide modes spread across several dimensions, we test along the direction that a tentative binary split proposes. Given a cluster C, let k-means with $k = 2$ return centroids $\boldsymbol{\mu}_0, \boldsymbol{\mu}_1$, and set the unit split direction and projected sample

$$\boldsymbol{u} = \frac{\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0}{\|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0\|}, \qquad y(s) = \boldsymbol{u}^\top \boldsymbol{z}(s), \quad s \in C. \tag{4}$$

The statistic is $\hat{D}_C = \operatorname{dip}(\hat{F}_C)$, where $\hat{F}_C$ is the empirical distribution function of $\{y(s)\}_{s \in C}$. Coupling the test to the split it must license makes the procedure sensitive to the bimodality k-means detects, while preserving its calibration under H_0: projecting a unimodal (Gaussian) cloud onto any fixed direction returns a unimodal sample, regardless of where k-means places the cut.

3.4 Effective-sample-size calibration

The null distribution of the dip depends on the sample size: for m *independent* draws from the least-favourable unimodal law (the uniform; Hartigan and Hartigan 1985), $\operatorname{dip}(F_m) = O_p(m^{-1/2})$, and we denote by $q_{1-\alpha}(m)$ its $(1-\alpha)$ quantile. With autocorrelated data the nominal size n_C overstates the information content: the empirical distribution of n_C correlated values fluctuates about the population distribution at the rate of

$$n_{\text{eff}} = \frac{n_C}{a} \tag{5}$$

independent values, the effective sample size of a field with decorrelation area a (Cressie 1993; Dutilleul 1993). We therefore calibrate the observed dip against the null at the effective, not the nominal, size: the test rejects unimodality of C, and thereby licenses a split, when

$$\hat{D}_C \;>\; q_{1-\alpha}\big(\lfloor n_{\text{eff}} \rceil\big), \tag{6}$$

with $\lfloor\cdot\rceil$ the nearest integer. The quantile $q_{1-\alpha}(\cdot)$ has no closed form and is obtained by Monte Carlo (dip of uniform samples of size $\lfloor n_{\text{eff}}\rceil$, cached per size bucket), so the calibration adds negligible cost. Autocorrelation thus raises the significance threshold (through a smaller n_{eff}) without discarding data: the statistic $\hat{D}_C$ is still computed from all n_C pixels, retaining full detection power.

The quantile $q_{1-\alpha}$ uses the *uniform* law, the least-favourable unimodal distribution (Hartigan and Hartigan 1985): it bounds the level for any unimodal within-class law, but is conservative when that law is the Gaussian of Assumption 1. A recall-oriented variant, **ESS-Dip-R**, calibrates instead against the Gaussian null, drawing the reference dips from $\lfloor n_{\text{eff}}\rceil$ standard-normal samples; the two are the distribution-free and the model-matched ends of one calibration. The Gaussian reference is less conservative and recovers more structure while empirically preserving unit specificity (Sections 4.3 and 5.3); we report both and take the conservative uniform null as the default.

3.5 Within-class decorrelation area by local range estimation

The calibration (5) requires the decorrelation area (2) of the *within-class* residual $\boldsymbol{\varepsilon}$, not of the between-class mosaic. Estimating it from the whole scene conflates the two: large contiguous classes inflate the apparent range, a grows, n_{eff} collapses, and the test loses all power, the field-scale rather than the within-class autocorrelation being measured. We avoid this conflation by estimating a *locally* over small tiles, the principle that also motivates declustering, where spatially clustered support is kept from dominating a statistic (Isaaks and Srivastava 1989; Deutsch and Journel 1998).

Partition D into square tiles $\{T_j\}$ of side t; most tiles fall *within* a single class, so their second-order structure reflects $\boldsymbol{\varepsilon}$ alone. For each tile we take the first principal component of $\{\boldsymbol{z}(s)\}_{s\in T_j}$, form its autocorrelation by the fast Fourier transform ($O(t^2\log t)$), and record the lag ℓ_j at which it first falls below $1/e$, averaged over the two axes. The within-class decorrelation area is the squared median range,

$$a = \big(\text{median}_j\ \ell_j\big)^2, \tag{7}$$

the median being robust to the minority of tiles that straddle a class boundary, and the square turning a correlation length into the area of one decorrelation patch. This $1/e$-range proxy leaves the geometric constant relating range to area implicit. A more explicit route fits, by weighted least squares, the best of an exponential, spherical and Gaussian variogram model (nugget c_0, partial sill c, range a_j) to each tile and uses its closed-form integral range

$$a_j \;=\; 2\pi\,\frac{c}{c_0+c}\int_0^\infty r\,\rho(r)\,dr \;=\; \kappa\,\frac{c}{c_0+c}\,a_j^2, \tag{8}$$

with $\kappa = 2\pi,\ \pi,\ 0.2\pi$ for the exponential, Gaussian and spherical models respectively (the radial integral contributing a_j^2, $a_j^2/2$ and $0.1\,a_j^2$ to the common angular factor 2π), the nugget fraction $c_0/(c_0+c)$ enlarging the effective sample size. This integral range is derived for the variance of the sample *mean*; applied to the dip, a functional of the empirical distribution, it over-corrects, and Section 4.5 finds it markedly more conservative than the ℓ^2 proxy, which we therefore retain as the default. An assumption-free alternative, the Gaussian-simulation reference, is taken up in Section 3.7.

3.6 Adaptive trend–residual separation

A smooth drift $\boldsymbol{m}$ is itself unimodal yet, left in place, enlarges ℓ and biases the test. We remove it adaptively. For each band b fit a low-order polynomial surface $\hat{m}_b(s) = \boldsymbol{\beta}_b^\top \boldsymbol{\phi}(s)$ by ordinary least squares, where $\boldsymbol{\phi}$ collects the monomials in the pixel coordinates up to degree two, and let

$$R^2_{\text{poly}} = \frac{1}{p}\sum_{b=1}^{p}\left(1 - \frac{\sum_s \big(z_b(s) - \hat{m}_b(s)\big)^2}{\sum_s \big(z_b(s) - \bar{z}_b\big)^2}\right) \tag{9}$$

be the average variance explained by the drift. A genuine large-scale trend produces R^2_{poly} close to one, whereas a discrete mosaic, not well represented by a low-order polynomial, produces a small value. We therefore test the residual $\boldsymbol{\varepsilon} = \boldsymbol{z} - \hat{\boldsymbol{m}}$ in place of $\boldsymbol{z}$ whenever

$$R^2_{\text{poly}} > \tau, \tag{10}$$

and test $\boldsymbol{z}$ unchanged otherwise. Removing the drift only when it is present protects the power of the test on trend-free scenes, where detrending would needlessly strip signal. Equation (1) together with (10) is the universal-kriging decomposition restricted to the regime in which the drift is statistically resolvable.

3.7 A Gaussian-simulation reference

The effective-sample-size calibration (6) is an approximation: it maps an autocorrelated sample of size n_C to an independent sample of size n_{eff} and reads the dip quantile there. Two caveats attend it. First, the integral range (8) is derived for the variance of the sample *mean*, whereas the dip is a functional of the empirical *distribution*, governed by the level-crossing indicators rather than the field itself. For a Gaussian field the median-level indicator correlogram is the arcsin transform $\rho_I(\boldsymbol{h}) = \frac{2}{\pi}\arcsin\rho(\boldsymbol{h}) \le \rho(\boldsymbol{h})$, a standard result of indicator geostatistics (Goovaerts and Jacquez 2004), so the decorrelation area appropriate to the dip is the *indicator* integral range $a_I = \iint \rho_I\, d\boldsymbol{h}$, strictly below the field integral range ($a_{\text{int}} \approx 1.45\, a_I$ on our fields). This is the geostatistical reason the mean-based integral range over-corrects (Section 4.5). The $1/e$ proxy $a = \ell^2$ lies in the same indicator-range regime, within which the recovered $\hat{K}$ is insensitive to the precise area, so we retain it; Section 3.9 confirms the resulting level is controlled across the operating range of n_{eff}. Second, it presumes isotropy. A reference that avoids both is the geostatistical *neutral model* (Goovaerts and Jacquez 2004): simulate, by the spectral (fast-Fourier) method (Dietrich and Newsam 1993; Lantuéjoul 2002), an ensemble of unconditional Gaussian realisations $\{\tilde{y}^{(b)}\}_{b=1}^{B}$ on the support of C whose covariance is the fitted variogram of the projection y, and reject unimodality when

$$\hat{D}_C \;>\; \text{Quantile}_{1-\alpha}\big\{\text{dip}(\tilde{y}^{(b)})\big\}_{b=1}^{B}. \tag{11}$$

Algorithm 1 ESS-calibrated modality test for K (single ensemble member)

Require: field $\boldsymbol{z}$ on D; level α; degree-2 drift threshold τ; tile side t; minimum effective size ν

if $R^2_{\text{poly}}(\boldsymbol{z}) > \tau$ **then** ▷ Eq. (9)–(10)
  $\boldsymbol{z} \leftarrow \boldsymbol{z} - \hat{\boldsymbol{m}}$ ▷ trend removal
end if
$a \leftarrow (\text{median}_j\, \ell_j)^2$ ▷ local $1/e$-range area, Eq. (7)
stack $\leftarrow \{D\}$; leaves $\leftarrow 0$
while stack $\neq \emptyset$ **do**
  pop C from stack; $n_{\text{eff}} \leftarrow n_C/a$
  if $n_{\text{eff}} < \nu$ **then**
    leaves $\leftarrow$ leaves $+1$; **continue** ▷ insufficient independent information
  end if
  $(\boldsymbol{\mu}_0, \boldsymbol{\mu}_1) \leftarrow$ 2-MEANS(C); $\boldsymbol{u} \leftarrow (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)/\|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0\|$
  $\hat{D}_C \leftarrow \text{dip}\big(\{\boldsymbol{u}^\top \boldsymbol{z}(s)\}_{s\in C}\big)$ ▷ Eq. (3)–(4)
  if $\hat{D}_C > q_{1-\alpha}(\lfloor n_{\text{eff}} \rceil)$ **and** both children have $n \geq a$ **then**
    push both 2-means children onto stack ▷ Eq. (6)
  else
    leaves $\leftarrow$ leaves $+1$
  end if
end while
return $\hat{K}$ = leaves

For a *single* homogeneity decision on a full rectangular domain, testing $K = 1$ against $K \geq 2$, this Monte Carlo test makes no independence approximation and accommodates anisotropic or nested structure, and we show in Section 4.5 that it decides correctly where the effective-sample-size shortcut requires its calibration constant. Two limitations keep it from supplanting (6) as the estimator, however. It is exact only up to the *plug-in* variogram, fitted from the same data; and when iterated over the recursion its per-node variogram must be fitted on the non-rectangular support of a candidate cluster, and the repeated tests incur multiple comparisons. We implement the resulting recursive estimator (Section 4.5): a direct version over-splits, a multiple-testing correction makes it competitive with (6) on controlled fields, but its plug-in variogram on irregular supports keeps it behind on real imagery, and at the cost of B simulations per node against the cached, near-linear cost of (6). We therefore retain (6) as the method and use (11) as an assumption-light check on the leading split.

3.8 Recursive partitioning

The components above are combined into a top-down recursion (Algorithm 1). Beginning with $C = D$, a cluster is split into its 2-means children whenever the calibrated dip test (6) is significant and both children retain at least one decorrelation patch ($n \geq a$, so that $n_{\text{eff}} \geq 1$); otherwise it becomes a leaf. The estimate $\hat{K}$ is the number of leaves. To absorb the variability of the k-means initialisation the recursion is repeated B times with independent seeds and the median of $\{\hat{K}^{(b)}\}_{b=1}^{B}$ is reported.

3.9 False-split control

The calibration (6) is designed so that, in the absence of discrete structure, the recursion does not split. We give conditions under which the per-node test controls its level asymptotically. Write (C1) for the assumption that on C the residual $\boldsymbol{\varepsilon}$ is a stationary Gaussian random field whose correlogram ρ is nonnegative and absolutely summable, $\sum_{\boldsymbol{h}} \rho(\boldsymbol{h}) < \infty$; (C2) for the projection direction $\boldsymbol{u}$ being fixed; and (C3) for the hypothesis that the projected empirical process obeys a functional central limit theorem, $\sqrt{n_C/a_I}\,(\hat{F}_C - \Phi) \Rightarrow \mathbb{G}$, a Gaussian process, with Φ the projected Gaussian law and $a_I = \iint \frac{2}{\pi} \arcsin \rho \, d\boldsymbol{h}$ the indicator integral range (Section 3.7).

Proposition 1 (Asymptotic false-split control)**.** *Under* (C1)–(C3)*, as C grows with ρ fixed and $n_{\text{eff}} = n_C/a \to \infty$ with a proportional to the indicator integral range a_I* (Section 3.7)*, the rejection probability of the test* (6) *tends to at most α. Consequently, if every node of the recursion satisfies* (C1) *(the scene is a single class), then $\mathbb{P}(\hat{K} = 1) \to 1$.*

Proof Under (C3) the projected empirical process converges, $\sqrt{n_C/a_I}\,(\hat{F}_C - \Phi) \Rightarrow \mathbb{G}$. The dip is 1-Lipschitz in the supremum norm, $|\text{dip}(F) - \text{dip}(F')| \le \sup_x |F(x) - F'(x)|$ (Hartigan and Hartigan 1985), so by the continuous mapping theorem $\sqrt{n_C/a_I}\,\hat{D}_C \Rightarrow \text{dip}(\mathbb{G})$; as Φ is unimodal the limiting population dip is zero and $\hat{D}_C = O_p\big((n_C/a_I)^{-1/2}\big)$. The independent dip quantile satisfies $q_{1-\alpha}(m) = O(m^{-1/2})$ (Hartigan and Hartigan 1985), so comparing $\hat{D}_C$ with $q_{1-\alpha}(\lfloor n_{\text{eff}} \rceil)$ at $n_{\text{eff}} \propto n_C/a_I$ caps the asymptotic rejection probability at α. The recursion visits finitely many nodes, and a union bound gives $\mathbb{P}(\hat{K} = 1) \to 1$. $\square$

Remark 1. Condition (C3) carries the analytic weight. It is the empirical-process counterpart, at the indicator integral range, of the effective-sample-size heuristic; functional central limit theorems of this form are established for the empirical processes of strongly mixing and of associated sequences and fields, and (C1) places $\boldsymbol{\varepsilon}$ in the latter class through Pitt (1982); Bulinski and Shashkin (2007). A complete verification for the lattice empirical process under our exact conditions is beyond the present scope, so we state (C3) as a hypothesis rather than derive it. The proportionality constant of the supremum functional is set through the ℓ^2 proxy (Section 3.5), which lies in the indicator-range regime (ℓ^2 of the order of a_I) and whose level Figure 1 confirms. Two further idealisations remain: the split direction is data-dependent in practice, (C2) being the fixed-direction relaxation under which Figure 1 shows no inflation; and non-Gaussian within-class laws fall outside (C1), where the distribution-free uniform null of ESS-Dip is the safer reference and ESS-Dip-R is not licensed.

The Gaussian-simulation reference (11) controls the level exactly for a single homogeneity test and so provides an assumption-light check on the leading split, but, as Section 4.5 shows, it does not yield a better recursive estimator. The effective-sample-size test (6) is thus an approximation whose adequacy is assessed empirically: Section 4 finds that, over the 75 null and trend realisations with no discrete classes, the procedure returns $\hat{K} = 1$ in every case, whereas the gap statistic and the silhouette, Calinski–Harabasz and Davies–Bouldin indices report spurious clusters in all of them.

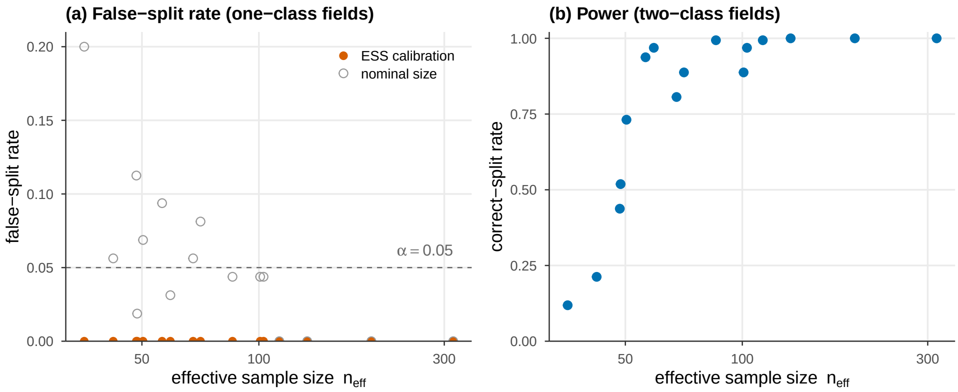

Fig. 1 Direct validation of the effective-sample-size calibration on a single homogeneity decision, 160 realisations per cell. (a) On one-class Gaussian fields the false-split rate of the calibrated test stays at 0.00, at or below the nominal $\alpha = 0.05$, across effective sample sizes from 35 to over 300; calibrating at the nominal size instead exceeds α (open symbols). (b) On two-class fields the power rises with n_{eff} to one, so the level control is not the trivial conservatism of a test that never rejects.

The calibration can be probed directly, in isolation from the recursion. On single-class Gaussian fields spanning effective sample sizes from 35 to over 300 (the regime the recursion visits), the per-node test of Section 3.4 never produces a false split: its empirical false-split rate is 0.00 at every n_{eff} (Figure 1a), at or below the nominal α, and unchanged whether the projection direction is fixed or chosen by 2-means, so the data-dependent direction does not inflate the level. Calibrating instead at the nominal size n_C breaks this control (open symbols). The conservatism is not vacuous: on two-class fields the power rises with n_{eff} to one (Figure 1b). The effective size borrowed from sample-mean theory therefore calibrates the dip across the operating range, holding the level conservatively, even though its exact constant for the dip functional is left undetermined.

3.10 Computational considerations

The range estimator (7) costs one fast Fourier transform per tile, $O(N \log N)$ in total. Each visited node runs one 2-means fit, of cost $O(n_C p)$ with mini-batch updates, and one dip evaluation, $O(n_C \log n_C)$ after sorting the projection; the Monte Carlo quantiles $q_{1-\alpha}(\cdot)$ are computed once per effective-size bucket and reused. As the recursion visits each pixel in $O(\hat{K})$ nodes, one ensemble member is near-linear in N, and the B members are independent and trivially parallel. The procedure therefore scales to full scenes; Section 4.6 reports timings confirming a near-linear empirical exponent and a memory footprint a small multiple of the data, together with a constant-factor advantage over the gap statistic, whose reference replicates multiply its cost.

4 Simulation study

We assess the procedure on synthetic scenes with known ground truth, where the number of classes, the strength of spatial autocorrelation, the presence of a trend, and the class separability are all controlled. The design isolates the two failure modes of Section 3.2, autocorrelation and large-scale drift, and lets us verify the false-split control of Proposition 1 directly, by counting how often each method recovers a single class when the scene contains exactly one.

4.1 Generative model

All scenes live on a 96×96 lattice with $p = 6$ bands. The elementary building block is a unit-variance Gaussian random field g_σ, obtained by convolving Gaussian white noise with an isotropic Gaussian kernel of bandwidth σ (which sets the autocorrelation range) and standardising. Four world types instantiate the decomposition (1):

Null ($K = 1$): each band an independent g_σ, $\sigma \in \{3, 6, 9\}$. Pure autocorrelation, no discrete structure.
Trend ($K = 1$): each band $\theta\, t(s) + 0.4\, g_3(s)$, with t a standardised planar–bilinear polynomial surface and trend amplitude $\theta \in \{1.5, 3.0\}$. Smooth large-scale variation, no classes.
Structured ($K \in \{2, 3, 4, 5\}$): a contiguous label map is drawn as the pixelwise arg max of K independent fields g_{σ_f}, $\sigma_f \in \{6, 9\}$; class k is assigned a mean spectrum $\boldsymbol{\mu}_k \sim \mathcal{N}(\mathbf{0}, \delta^2 I_p)$ with separation $\delta \in \{2, 3, 4\}$, and a per-band autocorrelated noise g_2 is added. Discrete classes embedded in autocorrelated noise.
Mixed ($K \in \{3, 4\}$): a structured scene ($\delta = 3$, $\sigma_f = 6$) plus a global polynomial trend of amplitude $\theta \in \{1.5, 3.0\}$. The realistic case: classes *and* drift together.

The factorial of these settings yields 33 cells (3 null, 2 trend, 24 structured, 4 mixed); each cell is realised 15 times with independent seeds, for 495 scenes in total.

4.2 Methods and metrics

We compare fifteen criteria. Four are standard internal indices evaluated against an independence null over $K = 1, \ldots, 8$: the gap statistic (Tibshirani et al. 2001) with its Tibshirani rule, the average silhouette, Calinski–Harabasz, and Davies–Bouldin. (The latter three cannot return $K = 1$ by construction, an asymmetry we return to below.) Two are the calibrated-validity-index method of Hennig and Lin (2015), the procedure ESS-Dip extends: the average silhouette width calibrated by parametric bootstrap against a null model, run both with an *i.i.d.* Gaussian null (the autocorrelation-blind application) and with a *spatial* null (phase-randomized surrogates preserving each band's variogram, their autocorrelation-aware variant adapted to the gridded setting); K is taken where the calibrated index, in standard-deviation units, first exceeds a significance threshold. A seventh baseline, plain *dip-means*, applies the same modality test at the nominal sample size, without the effective-sample-size calibration, the local range, or the trend removal, isolating the contribution of that spatial machinery. The eighth is the proposed method of Section 3, denoted **ESS-Dip**: Algorithm 1 with the local within-class range (7), adaptive trend removal (10), and a $B = 5$ ensemble; we fix

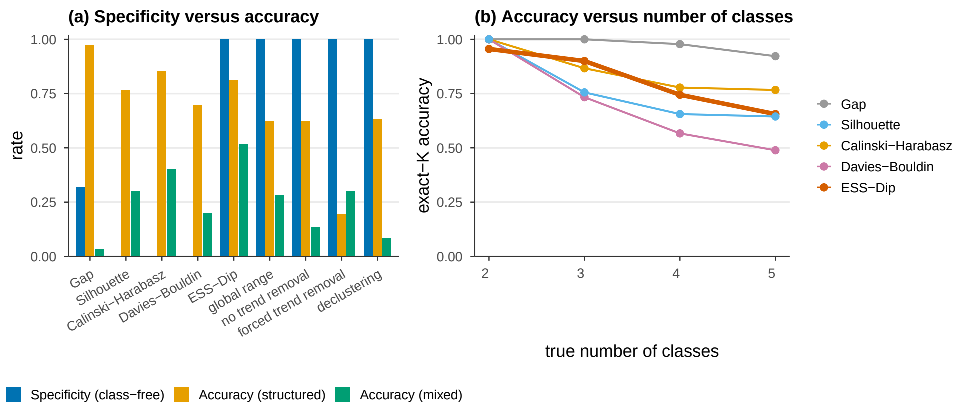

Fig. 2 Left: specificity (class-free scenes) against exact-K accuracy on structured and mixed scenes, by method. Only the proposed family attains unit specificity; among them ESS-Dip recovers the most structure. Right: exact-K accuracy versus the true number of classes on structured scenes. For legibility the panels show the four standard indices and the ESS-Dip family; the calibrated-validity-index and Gaussian-simulation methods of Table 1 are omitted.

$\alpha = 0.05$, $\tau = 0.25$, tile side $t = 24$, and $\nu = 12$ throughout, with no per-scene tuning. The ninth, **ESS-Dip-R**, is its recall-oriented variant: it calibrates the dip against the Gaussian null of Assumption 1 rather than the uniform least-favourable null, which is less conservative and recovers more structure at the same specificity (Section 4.3). The remaining four are ablations of ESS-Dip that disable one component each, to attribute the contribution of every ingredient: *global range* (the ℓ^2 range estimated over the whole image rather than as a tile median); *no trend removal* ($\tau = \infty$); *forced trend removal* ($\tau = 0$); and *declustering*, the precursor that subsamples the scene to one pixel per decorrelation patch and applies the dip test to the thinned, approximately independent sample. The fourteenth, *SGS-Dip*, replaces the per-node effective-sample-size calibration by the recursive Gaussian-simulation reference (11) of Section 3.7 ($B =$ 40 simulations per node); the fifteenth adds a Bonferroni multiple-testing correction to it.

Three metrics summarise performance. *Specificity* is the fraction of null and trend scenes (the 75 scenes with $K = 1$) for which $\hat{K} = 1$; it is the empirical analogue of Proposition 1. *Exact-K accuracy* is the fraction of scenes with $\hat{K} = K_{\text{true}}$, reported separately for the structured and mixed worlds, with the mean absolute error MAE $= \mathbb{E}\,|\hat{K} - K_{\text{true}}|$. The *balanced score* is the mean of specificity, structured accuracy and mixed accuracy, a single figure that rewards methods strong across all three regimes rather than in one.

4.3 Results

Table 1 reports the three metrics; Figure 2 displays the specificity–accuracy trade-off and the decay of accuracy with the true number of classes.

Table 1 Performance over the 495 synthetic scenes. Specificity is the $\hat{K}$=1 rate on the 75 class-free (null+trend) scenes; accuracy is the exact-K rate. Best in each column in bold.

Method	Specificity ($\hat{K}$=1)	Structured Acc.	Structured MAE	Mixed Acc.	Balanced score
Standard internal indices (independence null)					
Gap statistic	0.00	**0.97**	**0.03**	0.03	0.336
Silhouette	0.00	0.76	0.32	0.30	0.355
Calinski–Harabasz	0.00	0.85	0.18	0.40	0.418
Davies–Bouldin	0.00	0.70	0.41	0.20	0.299
Calibrated validity index (Hennig and Lin 2015)					
i.i.d. null	0.00	0.67	0.42	0.17	0.280
spatial null	0.80	0.72	0.38	0.17	0.561
Modality test without spatial calibration					
Dip-means (nominal)	0.52	0.92	0.09	0.02	0.486
Proposed					
ESS-Dip	**1.00**	0.81	0.35	0.50	0.771
ESS-Dip-R (recall)	**1.00**	0.89	0.19	**0.72**	**0.868**
Ablations of ESS-Dip					
global range	1.00	0.63	0.87	0.28	0.638
no trend removal	1.00	0.63	0.89	0.13	0.587
forced trend removal	1.00	0.19	1.95	0.30	0.498
declustering	1.00	0.63	0.83	0.08	0.572
Exact reference as a recursive estimator					
SGS-Dip	0.85	0.49	0.91	0.27	0.535
+ multiple-testing correction	0.99	0.79	0.26	0.35	0.710

Standard indices over-detect without exception.

On the 75 class-free scenes every standard index reports spurious structure: the $\hat{K}$=1 rate is 0.00 for all four. The failure is severe, not marginal: the gap statistic returns on average $\hat{K} = 7.7$ classes on fields that contain none, and the silhouette, Calinski–Harabasz and Davies–Bouldin indices average 2.9, 3.6 and 4.1. This is the over-detection anticipated in Section 3.2, now quantified: under spatial autocorrelation an independence null cannot recognise the absence of clusters.

The proposed family controls the false-split rate.

Every member of the proposed family, ESS-Dip and all four ablations, attains specificity 1.00: in 75 out of 75 class-free scenes it returns a single class. This is the empirical content of Proposition 1, and it holds whether the class-free scene is stationary (null) or carries a smooth drift (trend), confirming that the effective-sample-size calibration and the adaptive detrending together neutralise both failure modes.

Specificity does not cost competitiveness.

On the structured scenes the gap statistic is most accurate (0.97): with well-separated classes and an aligned null it is hard to beat, and we do not. ESS-Dip reaches 0.81, ahead of the silhouette (0.76), Davies–Bouldin (0.70) and close to Calinski–Harabasz (0.85), while being the *only* method with non-zero specificity. On the realistic mixed scenes ESS-Dip reaches 0.50, ahead of every standard index (0.40 for the best), because the adaptive detrending removes the drift that collapses them. The recall variant ESS-Dip-R does better still: calibrating against the Gaussian null of Assumption 1 rather than the conservative uniform one, it raises structured accuracy to 0.89 and mixed accuracy to 0.72 at the same unit specificity, for a balanced score of 0.868, the best of any method. ESS-Dip itself scores 0.771, already far above the next non-proposed method (0.638) and more than twice the best standard index (0.418). These synthetic fields are Gaussian, so the model-matched ESS-Dip-R is favoured here; its specificity and recall gains carry to the real scenes of Section 5. Standard indices that look strong on pure structured scenes collapse once specificity is taken into account.

Comparison with the method ESS-Dip extends.

The calibrated-validity-index framework of Hennig and Lin (2015) is instructive precisely because the present method generalises it. Applied with an independence (i.i.d.) null it inherits the over-detection of the standard indices (specificity 0.00, balanced score 0.280), confirming that the framework alone, without a spatial null, does not address the problem. With its autocorrelation-aware (spatial) null it recovers much of the lost specificity (0.80) and a comparable structured accuracy (0.72), a clear vindication of its central idea. Two differences remain, and they are the contribution of this paper. First, its specificity, at 0.80, still admits one class-free scene in five as clustered, where the within-class calibration of ESS-Dip admits none. Second, and decisively, on the mixed regime its accuracy is only 0.17, no better than the i.i.d. variant and far below ESS-Dip's 0.50, because a phase-randomized null preserves the trend along with the autocorrelation and so cannot separate a smooth drift from discrete classes. The explicit trend–residual decomposition of Section 3.6, absent from the calibrated-index framework, is what closes this gap. This comparison is insensitive to the significance threshold: the value used, $V > 1.96$ (the one-sided 2.5% level for the calibrated index under approximate normality), and the alternatives $V > 1.64$ and $V > 2.33$ all leave the conclusion intact: the i.i.d. null fails specificity (0.00) at every threshold, and the spatial null controls specificity (0.77–0.83) but not the mixed regime (≤ 0.08) at every threshold. Figure 3 makes the mixed-scene behaviour concrete on one realisation: the gap statistic and the Hennig–Lin null fragment the scene into six and eight clusters, whereas ESS-Dip recovers the three true classes.

4.4 Ablation

The ablations isolate where ESS-Dip's accuracy comes from (Table 1, lower block). The single largest contribution is the *local within-class range* of Section 3.5: replacing it by a scene-wide range drops structured accuracy from 0.81 to 0.63 and mixed accuracy from 0.50 to 0.28. This confirms the diagnosis of Section 3.5 (a global range conflates the between-class mosaic with the within-class scale, inflating ℓ, collapsing n_{eff}, and starving the test of power) while leaving specificity untouched. *Adaptive* trend removal matters in both directions: forcing detrending on every scene ($\tau = 0$)

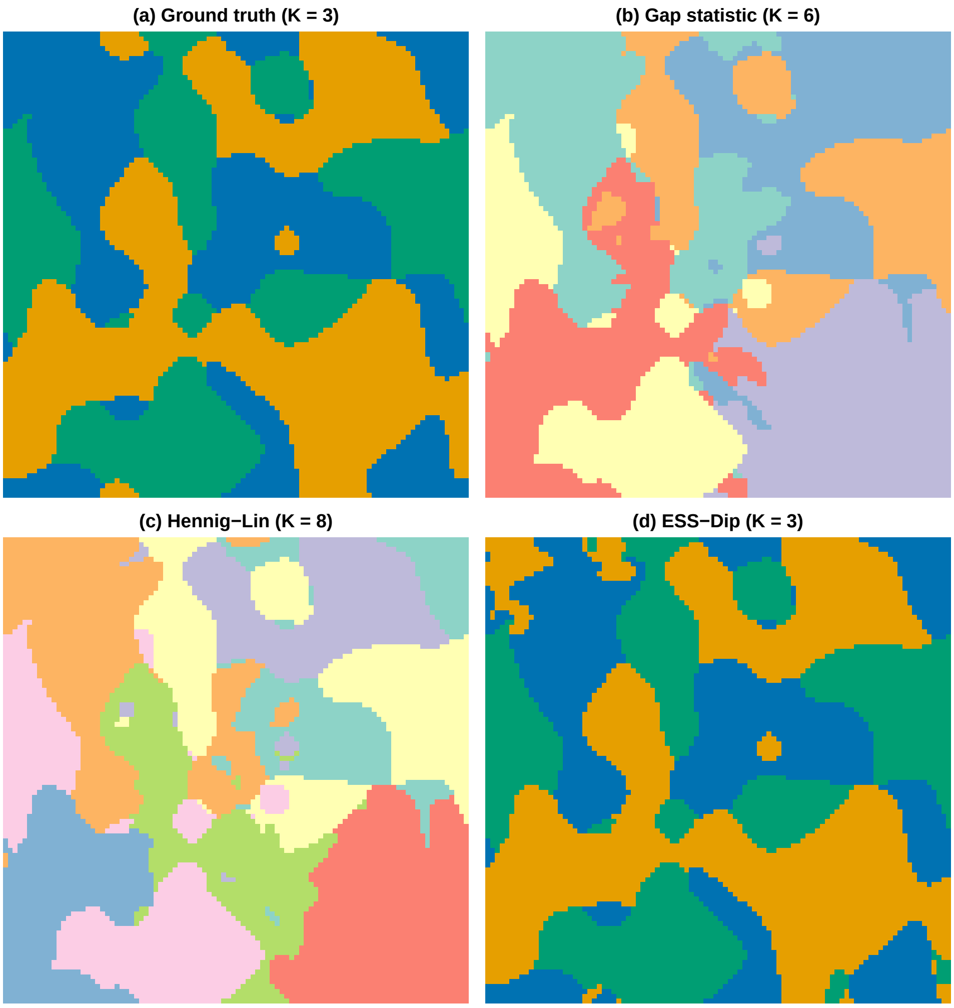

Fig. 3 Recovery on a representative mixed scene (three classes plus a smooth trend). The gap statistic over-segments into six clusters (b) and the Hennig–Lin spatial null into eight (c), both reading the trend and the within-class autocorrelation as extra structure; ESS-Dip removes the trend and calibrates to the effective sample size, recovering the three true classes (d), whose partition matches the ground truth (a). Cluster colours in (b) and (c) are arbitrary; in (a) and (d) they are matched, so a correct recovery reproduces the colours of panel (a).

destroys structured accuracy (0.19), because the polynomial absorbs genuine between-class signal when no trend is present; disabling it ($\tau = \infty$) halves mixed accuracy (0.13), because an unremoved drift inflates the range. Removing the drift *only when present* attains both (0.81 structured, 0.50 mixed). The declustering precursor, which discards all but one pixel per patch, matches the others on specificity but trails on accuracy (0.08 mixed), confirming the value of calibrating on the full pixel set rather

Table 2 The decorrelation-area constant and the exact null. $\hat{K}$ under the ℓ^2 proxy and under the variogram integral range; and the Gaussian-simulation homogeneity test (11) (p-value of the leading split, $B = 120$).

		$\hat{K}$		SGS null (11)
World	true K	$a = \ell^2$	integral range	p (decision)
Null	1	1	1	0.35 (K=1)
Trend	1	1	1	0.46 (K=1)
Struct	4	4	2	0.00 (K≥2)
Mixed	3	3	1	0.00 (K≥2)

than thinning it. At the opposite extreme, plain dip-means (Table 1) removes the spatial apparatus entirely and applies the dip at the nominal size: the modality signal alone recovers structure well (0.92, second only to the gap statistic), but without the effective-sample-size calibration its specificity falls to 0.52, and without trend removal its mixed accuracy collapses to 0.02. The contribution of ESS-Dip is therefore not the modality test, which is already strong, but the spatial calibration and trend separation that turn it from a high-recall, low-specificity splitter into a precision-controlled homogeneity test, at a modest cost in structured recall (0.92 to 0.81).

4.5 Geostatistical reference and the effective-sample-size approximation

Section 3.5 offers, alongside the default $1/e$-range proxy, a fitted variogram and its integral range, and Section 3.7 offers the exact Gaussian-simulation reference (11) in place of the effective-sample-size shortcut (6). Table 2 compares the three on one realisation of each world (six seeds, medians).

As a single homogeneity test the Gaussian-simulation reference (11) is exactly right: it accepts homogeneity on the class-free worlds ($p = 0.35, 0.46$) and rejects it on the structured and mixed worlds ($p < 10^{-3}$), with no effective-sample-size approximation. The constant κ also matters: the mean-based integral range (8) is some three times larger than ℓ^2 on these fields, so it is markedly more conservative, preserving specificity but losing recall (structured $\hat{K} = 2$, mixed $\hat{K} = 1$); the ℓ^2 proxy is, for the dip, the better-calibrated of the two scalar choices, consistent with the caveat of Section 3.7 that an integral range derived for the sample mean over-corrects a statistic of the empirical distribution, whose appropriate scale is the smaller indicator integral range.

It is tempting to promote the exact reference to the estimator by calibrating *every* node of the recursion against simulations on its support. A direct implementation (SGS-Dip, $B = 40$) over-splits: specificity 0.85, structured accuracy 0.49, balanced score 0.535 against ESS-Dip's 0.771 (Table 1). This has two causes. First, the per-node test is repeated across the recursion without correction, so the per-node level no longer controls the family-wise false-split rate. Second, away from the leading split a node is a spatially scattered subset whose variogram must be fitted on a masked, non-rectangular support, biasing the simulated null.

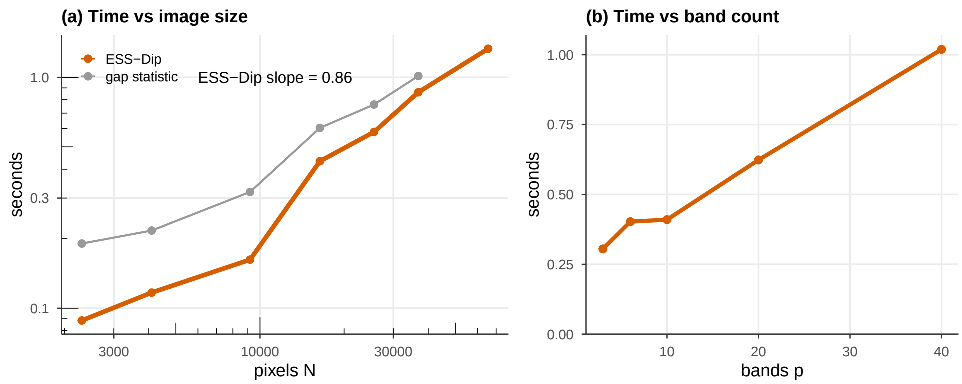

Fig. 4 Computational performance (single-threaded). (a) Wall-clock time versus pixel count N on a log–log scale, with the fitted ESS-Dip scaling exponent; (b) time versus band count p at $N = 128^2$.

The first cause is standard and correctable. A Bonferroni per-node level $\alpha/K_{\max}$, with the matching number of simulations, restores specificity to 0.99 and lifts the balanced score to 0.710, close to ESS-Dip's 0.771, the residual shortfall concentrated in the mixed regime (0.35 versus 0.50; SGS-Dip + correction, Table 1). So the simulation null is a viable recursive estimator once multiple testing is controlled. The second cause is more stubborn: on real imagery, where supports are irregular and non-stationary, even the corrected estimator trails ESS-Dip: single-class-window specificity 0.82 and 0.98 against 0.99 and 1.00 (Section 5.3), and over-detection of full scenes. The conclusion is therefore measured rather than categorical: the recursive simulation null becomes competitive on controlled fields once corrected, but its plug-in variogram on irregular supports, together with its cost of B simulations per node against the cached, near-linear effective-sample-size calibration, leaves (6) the better practical choice. We retain it as the method and use (11) as an assumption-light check on the leading split.

4.6 Computational performance

Figure 4 reports single-threaded wall-clock time on structured scenes. Across image sizes from $\sim$ 2,300 to $\sim$ 66,000 pixels, ESS-Dip scales with an empirical exponent $\mathrm{d}\log t/\mathrm{d}\log N = 0.86$ (panel a), near-linear, consistent with Section 3.10, and is a constant factor (about 1.2–2×) faster than the gap statistic, which scales similarly but carries the overhead of its reference replicates. Cost grows sub-linearly with the band count p (panel b: 0.31 s at $p = 3$ to 1.02 s at $p = 40$, at $N = 128^2$). Peak memory at $256 \times 256 \times 6$ is 8.9 MB, about three times the input cube, so the method is memory-light at scene scale. The exact Gaussian-simulation homogeneity test (11) with $B = 120$ costs 0.10–0.34 s over the same sizes, comparable to a single ESS-Dip evaluation, as it concerns one node; a fully recursive simulation null would multiply this by the number of nodes, which is the cost the effective-sample-size calibration is designed to avoid.

4.7 Robustness to anisotropy

The decorrelation area (7) is estimated from a radially averaged (isotropic) variogram, yet directional autocorrelation does not degrade the method. On null and structured worlds whose within-class range differs by a factor of up to eight between the two axes, ESS-Dip retains specificity 1.00 and structured accuracy 1.00 at every anisotropy ratio tested $(1, 2, 4, 8)$, whereas the gap statistic's specificity on the same class-free fields falls from 0.88 to 0.12 as the field elongates. The isotropic scalar is thus an adequate summary even under marked anisotropy, the modality signal it feeds being direction-agnostic, and a fully anisotropic variogram (Section 6) is a refinement rather than a necessity.

5 Application to land-cover imagery

We now turn to real scenes, where the question is no longer whether a method recovers a known K but whether it behaves sensibly on data with genuine spatial autocorrelation, smooth illumination gradients, and spectrally overlapping classes. We use three benchmarks spanning two sensing modalities and report two experiments: a direct test of the specificity property of Section 3.9 on regions known to contain a single class, and a full-scene estimate of the number of classes.

5.1 Data and preprocessing

Two airborne hyperspectral scenes with per-pixel ground truth are used: the *Indian Pines* (145×145, 200 bands after removal of water-absorption bands, 16 classes; Baumgardner et al. 2015) and the *Salinas* scene (512×217, 204 bands, 16 classes), both standard in the classification literature. Each cube is standardised per band and reduced by principal components to ten dimensions, retaining 95.5% and 99.6% of the variance respectively. As a multispectral counterpart, the common satellite setting, we use a 246×237 Sentinel-2 Level-2A scene over the Po Valley with ten surface-reflectance bands (B02–B12), labelled by the ESA WorldCover product (Zanaga et al. 2022); four classes exceed 1% of the scene (crop 83%, tree 9%, grass 5%, built 4%), so its nominal number of classes is four. Bands are standardised; no dimensionality reduction is needed.

5.2 Within-class autocorrelation

The calibration presumes a within-class autocorrelation that is real, estimable, and separable from the between-class mosaic; we verify all three on the data. Averaging the radial empirical semivariogram of the first principal component over pure single-class windows, each scene shows a well-defined within-class structure (Figure 5): an exponential model fits Indian Pines and Salinas (ranges 1.8 and 1.4 pixels) and a spherical model fits Sentinel-2 (range 5.4 pixels), with $1/e$ correlation lengths of two to three pixels. The indicator-to-field integral-range ratio derived in Section 3.7 holds on these real variograms: $a_{\text{int}}/a_I = 1.51$, 1.54 and 1.51 for the three scenes, close to the 1.45 of the Gaussian fields. Local estimation separates the within-class range from the scene scale: the tile-median range is 3 to 4 pixels, whereas the scene-wide estimate,

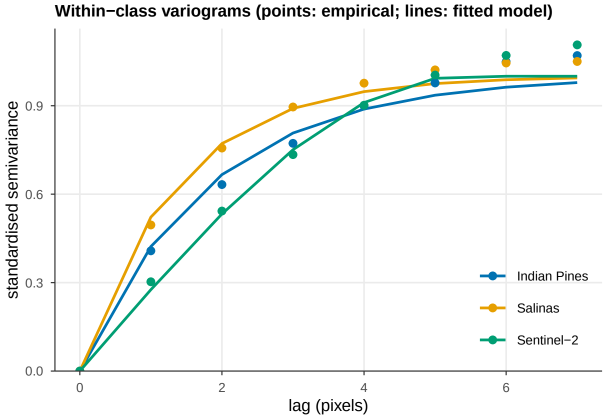

Fig. 5 Within-class variograms on the real scenes: the radial empirical semivariogram of the first principal component (points), averaged over pure single-class windows and standardised to unit sill, with the best-fitting exponential, spherical or Gaussian model (lines). The within-class autocorrelation the calibration relies on is real and has a well-defined range on actual imagery.

conflating the between-class mosaic, is three to fifteen times larger (14.5 to 46 pixels). The premise that most tiles fall within a single class holds on the contiguous scenes (62% of Salinas and Sentinel-2 tiles are single-class) and weakens on the fragmented Indian Pines fields (27%), where the tile median nonetheless recovers a within-class scale (3.5 against the 16-pixel scene range), the robustness for which the median is chosen.

5.3 Specificity on single-class regions

A region covered by a single land-cover class is the real-data analogue of the null and trend worlds of Section 4: its pixels are autocorrelated and spectrally variable (soil, moisture and illumination vary within a crop field), yet the true number of classes is one. We extract all 16×16 windows in which a single ground-truth class covers at least 90% of the pixels (67 windows across Indian Pines and Salinas, 60 in the Sentinel-2 scene, all of the dominant crop class), and record the number of clusters each method reports. Table 3 summarises the outcome; the proposed-family ablations of Section 4.2 all behave like ESS-Dip here (specificity ≥ 0.97) and are omitted for brevity.

The simulation result reproduces on real imagery. The silhouette, Calinski–Harabasz and Davies–Bouldin indices split a single real land-cover class into two or more sub-clusters in *every* window of both modalities; the gap statistic does so in all but the cleanest hyperspectral windows ($\hat{K}$=1 rate 0.64) and, on the multispectral crop fields, fragments a single class into nearly seven on average. ESS-Dip identifies the region as a single class in 99% of the hyperspectral and 100% of the multispectral windows. The within-field spectral variability that the standard indices read as structure is precisely what the effective-sample-size calibration discounts. The two closer baselines confirm the attribution on real data: the Hennig–Lin spatial null, calibrated for autocorrelation, improves on the standard indices over the hyperspectral windows (0.72) but collapses on the crop-dominated Sentinel-2 fields (0.07, mean $\hat{K} = 4.8$),

Table 3 Number of clusters reported on single-class windows (true $K = 1$). A method calibrated for spatial autocorrelation should return $\hat{K} = 1$.

	Hyperspectral (67 win.)		Sentinel-2 (60 win.)	
Method	$\hat{K}$=1 rate	mean $\hat{K}$	$\hat{K}$=1 rate	mean $\hat{K}$
Gap statistic	0.64	1.79	0.00	6.97
Silhouette	0.00	2.27	0.00	2.35
Calinski–Harabasz	0.00	2.70	0.00	3.20
Davies–Bouldin	0.00	3.37	0.00	2.62
Hennig–Lin (spatial)	0.72	2.36	0.07	4.77
Dip-means (nominal)	0.84	1.24	0.63	1.55
ESS-Dip	**0.99**	**1.01**	**1.00**	**1.00**
ESS-Dip-R (recall)	0.99	1.01	1.00	1.00

where within-field variation defeats its phase-randomized reference; plain dip-means, the modality test without the effective-sample-size calibration, reaches 0.84 and 0.63. Both trail ESS-Dip's 0.99 and 1.00, isolating the contribution of the spatial calibration on real imagery. The result is not an artefact of the preprocessing: on the Indian Pines windows ESS-Dip returns a single class in every window across principal-component truncations from five to thirty components, window sides from 12 to 24 pixels, and purity thresholds from 0.85 to 0.95 (for those combinations that yield windows). The recursive Gaussian-simulation estimator, even with the multiple-testing correction of Section 4.5, is less reliable here: it returns a single class in 0.82 of the hyperspectral and 0.98 of the multispectral windows, below ESS-Dip, because its per-node plug-in variogram degrades on real, non-stationary support, the residual effect that the correction does not remove.

5.4 Full-scene estimation

Applied to an entire labelled scene, the methods estimate the number of land-cover classes present (Table 4). No criterion approaches the nominal sixteen classes of the hyperspectral scenes: the labels include many small, spectrally similar agricultural classes that are not separable without supervision, and the best standard index reaches nine (the gap on Salinas) while the others return two to six. ESS-Dip is deliberately conservative, reporting two classes on Indian Pines and five on Salinas. On the crop-dominated Sentinel-2 scene, where a single class covers 83% of the pixels, ESS-Dip returns that one dominant class, whereas the gap reports five. The recall variant ESS-Dip-R recovers more where the structure is separable, reporting eleven classes on Salinas at the same single-class-window specificity (Table 3), while remaining conservative where it cannot help: two on the spectrally overlapping Indian Pines and one on the imbalanced Sentinel-2, which are limited by separability and dominance, not by the calibration's conservatism.

These full-scene figures should be read against the precision–recall trade-off of Section 4.3. Exhaustive recovery of many overlapping classes from a single scene is beyond every criterion considered, and the conservatism of ESS-Dip is the price of

Table 4 Estimated number of classes on full scenes; the nominal number of labelled classes is given in parentheses. The ≥ 20 entry marks the recursive simulation estimator reaching its recursion cap without terminating.

Method	Indian Pines (16)	Salinas (16)	Sentinel-2 (4)
Gap statistic	3	9	5
Silhouette	2	2	2
Calinski–Harabasz	2	6	3
Davies–Bouldin	2	6	2
ESS-Dip	2	5	1
ESS-Dip-R (recall)	2	11	1
SGS-Dip (corrected)	2	≥ 20	6

its specificity: a procedure that refuses to invent structure will, on a scene dominated by one class or composed of weakly separated ones, under-report rather than over-report. The recursive simulation estimator errs in the opposite direction, over-detecting to the imposed cap on Salinas and to six classes on the crop-dominated Sentinel-2 scene even after the multiple-testing correction, the plug-in variogram on the scene's irregular, non-stationary support compounding over a full-scene recursion. We regard the method's value as residing in that precision (it answers reliably whether genuine class structure is present beyond spatial autocorrelation) rather than in maximal class enumeration, and we return to this distinction, and to ways of relaxing it, in Section 6.

The number of classes is only half the picture; the quality of the induced partition completes it. Table 5 reports the adjusted Rand index (ARI) and normalised mutual information (NMI) of each labelling against the ground-truth map. Two things stand out. First, unsupervised recovery of these partitions is hard for every method: no ARI exceeds 0.52. Second, on the multi-class hyperspectral scenes ESS-Dip's partition quality trails the over-detecting gap statistic (Salinas ARI 0.32 versus 0.52), and on the crop-dominated Sentinel-2 scene, where it returns a single class, its ARI is zero. This is the partition-quality counterpart of the conservatism already visible in the class counts: a precision-oriented test that refuses to split does not, and is not meant to, maximise partition recovery. The numbers make the trade-off explicit and reinforce the reading of ESS-Dip as a homogeneity guard rather than a classifier.

6 Discussion

A precision-oriented criterion.

The results of Sections 4 and 5 draw a consistent picture across synthetic fields, hyperspectral scenes and multispectral imagery: standard internal criteria are powerful at recovering well-separated classes but cannot recognise their absence, whereas ESS-Dip controls the false-split rate at the cost of recall when classes are many or weakly separated. The two behaviours are not interchangeable failures of calibration but the two ends of a deliberate trade-off. By referring the dip statistic to the effective rather than the nominal sample size, the test is made conservative in exact proportion to the information the autocorrelation removes; the level α and the range ℓ are the knobs that set where on the precision–recall curve the method sits. We have fixed them once, without

Table 5 Partition quality on the full scenes: adjusted Rand index (ARI) and normalised mutual information (NMI) against the ground-truth land-cover map, with the number of classes each method returns. Computed on labelled pixels.

Scene	Method	$\hat{K}$	ARI	NMI
Indian Pines	Gap statistic	3	0.21	0.36
	Silhouette	2	0.18	0.37
	ESS-Dip	2	0.18	0.37
Salinas	Gap statistic	9	**0.52**	**0.71**
	Silhouette	2	0.18	0.37
	ESS-Dip	5	0.32	0.57
Sentinel-2	Gap statistic	5	0.10	0.14
	Silhouette	2	0.36	0.22
	ESS-Dip	1	0.00	0.00

per-scene tuning, and the resulting operating point is one of high precision: ESS-Dip answers reliably whether discrete structure is present *beyond* spatial autocorrelation, and is correspondingly cautious about enumerating structure that the data do not independently support. For the analyst this is the useful direction of error, a guard against the over-segmentation that Section 5.3 documents for the standard indices, but it is a genuine limitation when an exhaustive class inventory is the goal. This fixes the method's natural role: not a stand-alone classifier (on real multi-class scenes it under-reports), but a calibrated homogeneity test, a guard placed before an unsupervised classification or a stopping rule for a region-growing or regionalization procedure (Section 2.4), where a negative one can trust is worth more than an exhaustive but unreliable class count.

Relation to geostatistical null models.

Assumption 1 replaces the independence reference of the classical criteria by a unimodal Gaussian random field, the continuous-field counterpart of the spatial neutral models used in disease mapping (Goovaerts and Jacquez 2004) and of Moran spectral randomization (Wagner and Dray 2015). We realise it implicitly, through the effective-sample-size calibration of the dip, rather than by simulating reference fields. A more explicit route would calibrate the statistic against an ensemble of sequential-Gaussian-simulation realisations conditioned on the empirical variogram; this would relax the isotropy built into our scalar range ℓ and accommodate anisotropic or nested autocorrelation structures at the cost of the simulation effort. We see the present scalar calibration and a full simulation-based null as two points on a spectrum trading fidelity against computation, and the near-linear cost of the former (Section 3.10) as its principal practical advantage on megapixel scenes.

Limitations.

Three limitations bound the method's applicability. First, when many spectrally overlapping classes coexist, as in the sixteen-class hyperspectral scenes, no unsupervised criterion, ours included, approaches the nominal class count (Section 5.4); recovering such inventories requires supervision or auxiliary information that an internal criterion cannot supply. Second, strong class imbalance degrades recall: when one class dominates the scene, as the crop class does in the Sentinel-2 image, the leading split direction is governed by that majority and minority classes are not resolved. Third, the false-split control of Proposition 1 is an asymptotic statement resting on the empirical-process hypothesis (C3) and a fixed projection, not a finite-sample guarantee, and the 2-means direction it uses, though effective in our experiments, is a heuristic rather than an optimal choice.

Extensions.

Each limitation suggests a direction. The recall-oriented variant ESS-Dip-R realises one: trading the distribution-free uniform null for the model's Gaussian one recovers more structure at the same specificity (eleven classes on Salinas against five). Further recall could come from revisiting leaves at a relaxed level, or testing minority modes against the residual of the majority, to mitigate the imbalance that bounds the Sentinel-2 result. Replacing the scalar range by an anisotropic variogram model, and the uniform dip null by conditional simulation, would extend the calibration to non-stationary autocorrelation, although Section 4.7 shows the isotropic estimate is already robust to directional autocorrelation up to an eight-fold anisotropy ratio. The same effective-sample-size correction applies, unchanged, to spatio-temporal stacks, where the decorrelation "area" becomes a space–time volume. Finally, because our contribution is a stopping rule rather than a partitioning constraint, it composes naturally with spatially constrained clustering (Section 2.4): the calibrated test could decide when a regionalization has produced as many regions as the data independently support.

7 Conclusions

Selecting the number of clusters on spatial data is compromised by an assumption that the standard criteria make and that spatial data violate: that observations are independent. We have shown that the violation is not benign. Under spatial autocorrelation the gap statistic and the silhouette, Calinski–Harabasz and Davies–Bouldin indices over-detect without exception, inventing classes on featureless fields and fragmenting single land-cover regions on real imagery.

We have proposed a remedy framed in geostatistical terms. Recast as a calibrated test of cluster homogeneity, asking whether a region carries multimodality in excess of spatial autocorrelation, ESS-Dip evaluates the Hartigan dip along the locally optimal split direction and calibrates it against the *effective* sample size, estimated from a within-class autocorrelation range obtained by local range estimation, after removing a low-order trend whenever one is resolvable. The procedure carries an asymptotic false-split-control property and runs in near-linear time. On a factorial simulation with known ground truth it reaches unit specificity, and its recall variant ESS-Dip-R attains the best balanced score of fifteen criteria at the same specificity; on the Indian Pines, Salinas and Sentinel-2 scenes both identify single land-cover regions as homogeneous where the standard criteria do not. Their reliable negatives suit the family to use as a homogeneity guard or a stopping rule for region-growing as much as to cluster-number selection, spanning a precision–recall axis from the conservative distribution-free null to the higher-recall Gaussian one.

The broader message is that number-of-clusters selection on autocorrelated data must be calibrated to the information the data actually contain, not to their nominal size, and that the signal distinguishing classes from autocorrelation is modality, not dispersion. The effective sample size and declustering, long-standing tools of geostatistics, supply exactly what is needed to make a classical clustering criterion honest about spatial dependence. All code and data required to reproduce the simulation study and the applications are released.

Code and data availability

All code required to reproduce the simulation study and the applications (the synthetic-scene generators, the implementation of ESS-Dip and of every baseline and ablation, the benchmark driver, and the figure scripts) is openly available at https://github.com/franciscoparrao/ess-dip, released under the MIT licence, and will be archived on Zenodo with a permanent DOI on acceptance. The Indian Pines and Salinas hyperspectral scenes are the standard public benchmarks (Baumgardner et al. 2015); the Sentinel-2 Level-2A imagery is produced by the Copernicus programme and was accessed through the Microsoft Planetary Computer; the land-cover reference is the ESA WorldCover product (Zanaga et al. 2022). The synthetic scenes are generated deterministically by the released code from the seeds reported in the repository.

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